DocQuorum: controller, readers and tools

Automatic execution • bounded adaptive reader • one selected JSON

1 Inputs
- PDF / image
- Schema / template
- Optional task

Input evidence

2 Prepare & route
PYTHON CONTROLLER
Normalize input; inspect text, pages and layout
Run MinerU for scans when available
Pillow • Poppler • layout helpers • MinerU

Prepared evidence

3 Run reader(s)
MODEL CALLS
P / M / O / L
P M O L
See reader components below

Candidate JSONs

4 Check each candidate
DETERMINISTIC CODE
Schema + source evidence
Repair that candidate only
No model call

Checked candidate JSONs

5 Select one
DETERMINISTIC CODE
Route policy below
Quorum when needed: compare pairs; highest mean agreement
Keep one whole candidate

One JSON

6 Return
JSON + selection report
Restore template shape if supplied

Reader components: who reads, what they use, what they produce

Same adaptive reader agent, two evidence modes

Primary reader (P)
Adaptive agent
P
Evidence: PDF text + permitted page images
Tools: Poppler text/rendering; schema + chunk helpers; extraction-model API
Output: candidate P

Parse-based reader (M)
Same adaptive agent
M
Evidence: cached MinerU text / HTML; images when the plan permits
Tools: MinerU cache; schema + chunk helpers; extraction-model API
Output: candidate M

Separate runs and candidate state; shared source evidence, no shared answers.

Scheduled workflow

Column reader (L)
L
Detect columns → crop → read top to bottom
Tools: Poppler rendering; column detector / cropper; extraction-model API
Output: layout candidate L

Scheduled workflow

Ruled-table reader (L)
L
Detect ruled rows → stack header + row band → read fixed row count
Tools: Poppler rendering; row-band detector; extraction-model API
Output: layout candidate L

Scheduled workflow

One-shot reader (O)
O
Whole document in one extraction call
Tools: ExtractBench provider; page rendering; extraction-model API
Output: candidate O

Tools are called by Python, not chosen autonomously by the LLM

Poppler [CODE] → page count • PDF text • page images
MinerU parser [MODEL] → cached text blocks + HTML tables
Schema / row / layout helpers [CODE] → scopes • chunks • crops • row counts
Extraction model via API [MODEL] → schema-guided JSON replies

Inside P / M: bounded agentic loop

1 **PLAN [code]** — Choose eligible scope
2 **ACT [model]** — Read selected evidence
3 **OBSERVE [code]** — Truncated? Missing keys? Empty despite row evidence?
4 **Retry needed?** — Yes → 5 ADAPT [code]: Increase budget; require keys; split or reduce scope
No → Return candidate
Observed reply changes next call

Scopes: whole document / schema groups / text windows / scalar + list chunks

Python holds state and enforces retry rules; the LLM extracts values.

Which readers run, and how is one kept?

Document route	Readers attempted	Selection
Specialized layout	Column OR ruled (L)	Keep L
Text available	Primary (P)	Keep P
No text, ≤12 pages	P + M* + O	One → keep; multiple → quorum
No text, >12 pages	P then M*	Prefer retained M; otherwise P

Check chains (deterministic code)
- **P:** repairs → output checks
- **M:** spacing → repairs → output checks
- **L / O:** output checks

* M needs cached parse. Successful paths shown.

All stages after input submission are automatic. Agentic behavior is the bounded P/M feedback loop. No answer keys; no field-level candidate fusion.

1 Start with a real document

INPUT / OKLAHOMA COUNTY PURCHASE ORDER


Oklahoma County
Requisition Purchase Order

Purchase Order

"22601904"

Purchase Order Number 22601904

Purchase Order Date 08/28/2025

Department Emergency Management

I approve the processing of this purchase request and the encumbering on the designated account for a maximum stated herein.

Bill To: OKC CO EMERGENCY MANAGEMENT 320 RIO C T KARR OKLAHOMA CITY, OK 73102

Ship To: OKC CO EMERGENCY MANAGEMENT 320 RIO C T KARR OKLAHOMA CITY, OK 73102

Vendor: AVL SYSTEMS DESIGN LLC 14951 BRISTOL PARK BLVD EDMOND, OK 73013

VENDOR PHONE NUMBER	REQUISITION NUMBER	REQUISITIONER NAME	BUYER NAME	DELIVERY REFERENCE
405-748-1666	12602360	Nick Rutledge	Nick Rutledge	

NOTES

Sales Contact #OKHY2VAV - EOC Audio-Visual Equip

Pre-Alarm Purchase Order Number Must Appear On All Correspondence - Packing Sheets And Bills Of Lading

ITEM #	ORDERED	ITEM	QUANTITY	UOM	UNIT PRICE	EXTENDED PRICE
1	Statewide Contract # 3NY0219V - Emergency Operations Center (EOC) audio-visual equipment	1.0000	EACH	$6,721.8000	$6,721.80
	SL-# 12616666 - 80010			$6,721.80	

In accordance with 19 O.S. § 1500, the vendor shall furnish an invoice stating the terms and conditions of the vendor which must also be submitted therewith. Failure to furnish the invoice may result in delay in payment. Any violation of vendor shall, three attempts, the total price, the full purchase price, and the date of the purchase.

Requisitioning Officer: I approve the payment of the amount shown above, or as much thereof as may be allowed pursuant to law. Oklahoma County Purchasing Agent: I hereby sign this purchase order and encumbrance of the purchase order.

Oklahoma County Clerk: I hereby certify that the amount of the encumbrance has been entered against the designated appropriation account and balance and that sufficient funds remain unencumbered for the payment of this obligation.

Approver Name: Lonny Tate — Approval Date: 08/28/2025

Monique Priest — Deputy County Clerk

Kelly McMichael — Purchasing Agent

Purchasing Copy

Total Est. Price $6,721.80

Purchase Order Total $6,721.80

What the document contains

PO number	**22601904**
Date	28 August 2025
Vendor	AVL Systems Design LLC
Ordered items	One audiovisual-equipment line
Total	**$6,721.80**

What we want back

A structured object: PO number, vendor, line items, prices and total. The schema tells the reader which fields to extract.

The document supplies the evidence. The schema defines the answer's structure.

Source: ExtractBench short / oklahoma_county_purchase_order_avl_systems. Original PDF bundled in examples/.

2 Tell the reader what to extract

INPUT / SCHEMA OR TEMPLATE + OPTIONAL TASK

Schema: selected actual field types

```
{
  "type": "object",
  "properties": {
    "po_number": {
      "anyOf": [
        {"type": "string"},
        {"type": "null"}
      ]
    },
    "line_items": {"type": "array"}
  }
}
```

The complete schema also describes every field and each line-item property.

Alternative: a template

```
{
  "po_number": "",
  "line_items": [{
    "description": "",
    "quantity": "",
    "unit_price": ""
  }]
}
```

Optional task

"Extract every ordered item. Exclude totals and blank filler rows."

Source: Left: excerpt of the real PO schema. Right: teaching template and task, not inputs to the archived benchmark run.

3 First ask: what evidence is available?

PREPARE / THE SAME PO AS A DIGITAL PDF AND A DEGRADED SCAN

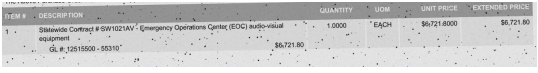
Digital PDF: usable embedded text



ITEM #	DESCRIPTION	QUANTITY	UOM	UNIT PRICE	EXTENDED PRICE
1	Statewide Contract # SW1021AV - Emergency Operations Center (EOC) audio-visual equipment GL #: 12515500 - 55310 $6,721.80	1.0000	EACH	$6,721.8000	$6,721.80

4,995 embedded characters.
One primary reading, using text and page images when the plan permits.

Degraded scan: no embedded text



ITEM #	DESCRIPTION	QUANTITY	UOM	UNIT PRICE	EXTENDED PRICE
1	Statewide Contract # SW1021AV - Emergency Operations Center (EOC) audio-visual equipment GL #: 12515500 - 55310 $6,721.80	1.0000	EACH	$6,721.8000	$6,721.80

0 embedded characters.
At one page: primary, cached-parse and one-shot readings can be compared.

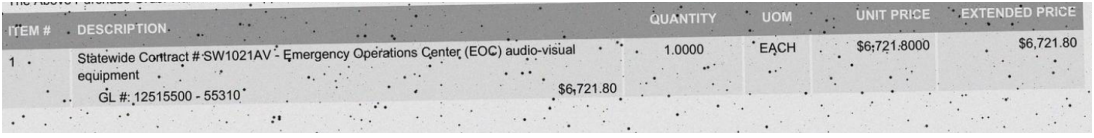
A scan can still contain OCR text. The real Texas W-14 example contains 931 embedded characters, so it follows the text-available route. Appearance alone does not choose the route.

Layout is checked first; otherwise embedded text and page count determine the route.

Source: Archived front-end measurements from the real digital PO, its benchmark-corrupted derivative, and Texas W-14 form.

4 MinerU turns the table into readable evidence

PARSE / ACTUAL TABLE CROP AND CACHED MINERU OUTPUT



ITEM #	DESCRIPTION	QUANTITY	UOM	UNIT PRICE	EXTENDED PRICE
1	Statewide Contract # SW1021AV - Emergency Operations Center (EOC) audio-visual equipment GL #: 12515500 - 55310	1.0000	EACH	$6,721.8000	$6,721.80
				$6,721.80	

Cached HTML excerpt

```
<tr><td>1</td>
<td>Statewide Contract # SW1021AV ...</td>
<td>1.0000</td><td>EACH</td>
<td>$6,721.8000</td>
<td>$6,721.80</td></tr>
```

Reader maps cells to the schema

```
"quantity": 1,
"unit_price": 6721.8,
"extended_price": 6721.8
```

The parse is text and HTML, not the final task JSON. The reader still has to interpret it.

Source: Actual cached MinerU parse for the degraded PO; description shortened with ellipsis. Full unedited parse is bundled.

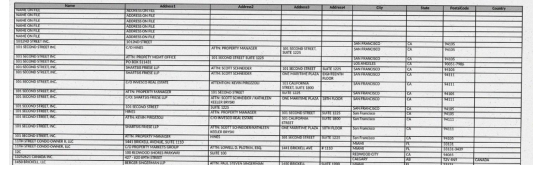
5 Change the view when layout defines the records

READ / TWO REAL SPECIALIZED ROUTES

Column flow: Oklahoma register



Ruled table: FTX scanned list



Stack the header above a band of up to 12 rows. Require one output entry per detected row.

Seven columns per page. Crop a column. Read top to bottom. Join continuation fragments.

Give the model the region that defines a record: a column or a counted row band.

Source: Real source-page excerpts: Oklahoma unclaimed-property register, p.2; corrupted FTX creditor list, p.2. Schemas and page PDFs bundled.

6 An incomplete reading can trigger another model call

READ / A REAL REQUIRED-FIELD RETRY

VENDOR PHONE NUMBER	VENDOR EMAIL	VENDOR NUMBER	REQUISITION NUMBER	REQUISITIONER NAME	BUYER NAME	DELIVERY REFERENCE
405-749-1866		5421	12602080	Nick Rutledge	Nick Rutledge	

First stored reply

```
"po_number": "22601904"

requisition_number: <absent>
```



Required-field re-read

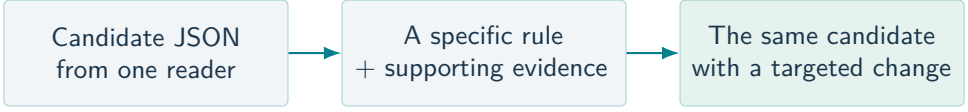
```
"po_number": "22601904",
"requisition_number": "12602080"
```

The code observes a missing key and requests another model read: a bounded agentic loop.

Source: Archived digital-PO first reply and second reply; the printed requisition number is shown above.

7 What does a deterministic check actually do?

CHECK / ONE CANDIDATE AT A TIME



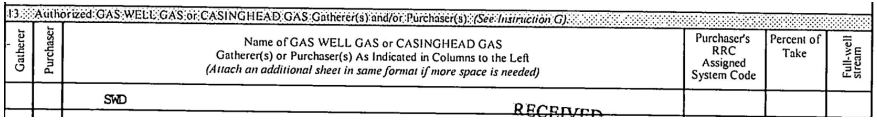
Example	What authorizes the change?	What changes?
Blank checkbox	Explicit schema convention	`null` becomes `false`
Rounded number	A uniquely supported printed value	Restore printed precision
Copied record	Matching values across two arrays	Remove the extra copy

A check applies only when its conditions match. It does not certify the whole answer.

Source: Current docquorum/checks implementation. These three rules are demonstrated with real inputs on the next pages.

8 Check example: blank means false, not unknown

SCHEMA-DRIVEN / TEXAS P-4 FORM, ITEM 13



13. Authorized GAS WELL GAS or CASINGHEAD GAS Gatherer(s) and/or Purchaser(s): *(See Instruction C)*:

Gatherer	Purchaser	Name of GAS WELL GAS or CASINGHEAD GAS Gatherer(s) or Purchaser(s) As Indicated in Columns to the Left *(Attach an additional sheet in same format if more space is needed)*	Purchaser's RRC Assigned System Code	Percent of Take	Full-well stream
		SWD RECEIVED			

The "Gatherer" cell for **SWD** is blank. The schema explicitly says:

"True if the Gatherer checkbox on this row is marked, false if it is blank."

MODEL CANDIDATE	→	AFTER THE CHECK
`gatherer: null`		`gatherer: false`

Why this rule applies: the field is boolean, its description defines blank as false, and it gives no separate meaning to null.

The check uses the schema convention. It does not look at the image or re-run OCR.

Source: Real P4-83-211_35 PDF and stored prediction. Current blank_boxes.apply replayed without model calls; complete artifacts bundled.

9 Check example: keep the precision printed in the table

TEXT-GROUNDED / BYLINE BANCORP, Q2 2023

(dollars in thousands, except per share data)	June 30, 2023
Income Statement	
Net interest income	$ 76,166
Provision (recapture) for credit losses	5,790
Non-interest income	14,291
Non-interest expense	49,328
Income before provision for income taxes	35,339
Provision for income taxes	9,232
Net income	26,107
Dividends on preferred shares	—
Net income available to common stockholders	$ 26,107

Original table: **dollars in thousands**. Net income: **26,107**.

The schema requests millions

The candidate says **26.11**. That is a rounded version of the source value. Printed 26,107 thousand = **26.107 million**.

Before `net_income: 26.11`

After `net_income: 26.107`

The rule restores a uniquely supported printed value; it does not estimate a new number.

Source: Byline Bancorp investor deck, physical p.18; actual stored candidate. printed_numbers.apply replay with text extracted from the source PDF.

10 Check example: one check was put into two lists

STRUCTURE / PUEBLO SCHOOL DISTRICT CHECK REGISTER

Check Register Summary — Pueblo School District 0[illegible]

Batch Year: 26 Bank: 01 Date Range: 10/01/2025 - 10/31/2025

Bank	Check	Type	Date	Vendor	Vendor Name	Amount
01	01033982	C	10/01/2025	651630	ALGEBROS LLC, THE	700.88
01	01033983	C	10/01/2025	50836	THE BUGMAN PEST CONTROL SERVICES	3,070.00
01	01033984	C	10/01/2025	45874	CAPLAN & EARNEST LLC	53,180.00

The printed row labels **01033982 as a check number**, with amount **700.88**.

Keep the actual check

```
checks[]:
  check_number: "01033982"
  amount: "700.88"
```

Remove its copy from batches

```
batches[]:
  batch_number: "01033982"
  amount: "700.88"
```

Why it applies: distinctive values match under different keys. At least three rows, and at least half of the judged batch rows, repeat this pattern.

562 batch entries become 51. All 511 check records stay unchanged.

Source: Real Pueblo register and stored prediction. copied_rows.apply replayed; the rule compares arrays, not the page image.

11 Checks do not combine the different readings

CANDIDATES / BACK TO THE DEGRADED PURCHASE ORDER

Selected field	**Primary**	**Parse-based**	**One-shot**
Requisition number	12602080	`null`	12602080
Address excerpt	BRISTOL PARK BLVD	BRISTOL PARK BLVD.	BRISTOL PARK BLVD.
Description excerpt	GL #: 12515500	GL #: 12515500	GL #,12515500

These are excerpts from **actual checked candidates**. Each reader used the source independently.

The parse candidate omitted the requisition key. The shape check added `null`; it did not recover the printed number.

Deterministic changes can improve consistency without fixing every missing or incorrect value.

Source: Archived checked outputs for oklahoma_county_purchase_order_avl_systems_corrupted. Full candidate JSONs are bundled.

12 Quorum keeps one complete candidate

SELECT + RETURN / THE STORED PURCHASE-ORDER DECISION

Mean agreement with the others

Primary	0.9348
Parse-based	0.9348
One-shot: selected	**0.9565**

Compare all candidate pairs. Select the highest mean agreement. Ties follow reader order.

Returned JSON: excerpt of one-shot


```
{
  "po_number": "22601904",
  "requisition_number": "12602080",
  "total_amount": 6721.8,
  "currency": "USD"
}
```


No fields are borrowed from the other candidates.

Agreement is not accuracy: in this real example, primary scored better against the answer key.

Source: Archived agreement.json and final.json. Selection sees candidates and schema, not the answer key. No new model inference.

95.43 OVERALL / 3RD IN THE PINNED SNAPSHOT

#	System	Short	Med.	Long	Overall
1	LlamaExtract Agentic Plus	96.94	94.45	90.18	95.91
1	Pulse (Effort)	96.46	95.01	93.51	95.91
3	DocQuorum (ours)	96.34	94.11	90.41	95.43
4	LlamaExtract Agentic	96.04	92.11	91.83	94.77
5	Codex (GPT-6 Sol Evidence)	95.99	92.42	87.25	94.57
6	Codex (GPT-5.6 Sol Evidence)	96.04	90.19	82.73	93.77
7	Codex (GPT-5.5)	95.68	91.15	78.88	93.57
8	Codex (GPT-5.5 Evidence)	95.59	88.72	87.97	93.35
9	Pulse (Non-effort)	93.17	92.22	89.07	92.70
10	LlamaExtract Cost-Effective	94.03	89.51	83.73	92.27
11	Codex (GPT-5.6 Terra Evidence)	95.60	86.69	77.36	92.26
12	OpenAI GPT-6 Astra	97.22	90.56	31.70	91.91
13	Codex (GPT-5.6 Luna Evidence)	93.50	87.43	80.05	91.16
14	Codex (GPT-6 Luna Evidence)	93.09	87.71	80.88	91.01
15	Reducto Deep Extract	94.20	80.47	92.01	90.44
16	Qwen3.8 Flash Next	94.82	87.81	37.74	89.88
17	Qwen3.8 27B	94.68	87.54	38.45	89.75
18	Extend (Max Context)	92.32	78.75	90.46	88.62
19	Qwen3.6 35B	92.86	86.37	36.90	88.11
20	Qwen3.5 35B	93.52	85.27	31.73	88.00
21	DeepSeek V4.1 Flash	94.44	81.49	22.23	87.11
22	Claude Code (Opus 4.8)	90.08	79.21	88.07	87.09
23	Claude Code (Opus 4.8 Evidence)	90.54	79.17	77.94	86.84
24	Qwen3.5 9B	92.31	81.12	32.22	86.10
25	Claude Code (Opus 5.5 Evidence)	89.63	77.96	80.83	86.06
26	Extend Extract	89.39	84.14	47.21	85.72
27	Datalab (Accurate + Balanced)	89.40	85.08	42.04	85.70
28	Reducto Extract	92.97	69.59	30.65	83.41
29	Kimi K3	94.64	69.64	4.99	83.17
30	Qwen3.5 4B	89.06	76.64	27.31	82.43
31	NuExtract3	86.41	77.01	32.71	81.02
32	GLM-5.3 Flash	96.30	51.56	27.83	80.75
33	Google Gemini 3.8 Flash	88.14	72.23	28.70	80.71
34	Google Gemini 3.5 Flash	87.87	69.76	27.90	79.84
35	Lift Datalab 9B	87.64	65.69	24.78	78.43
36	OpenAI GPT-5.4 Nano	77.43	76.37	35.81	74.90
37	Gemma4 26B	77.70	59.61	28.88	70.27
38	Gemma4 E4B	81.10	48.93	26.82	69.64
39	InternVL3.5 14B	80.70	39.04	19.41	66.35
40	InternVL3.5 8B	79.26	36.62	17.57	64.63
41	Qwen3.5 2B	71.22	50.93	25.93	63.40
42	Kimi-VL A3B	69.86	34.61	18.94	57.77
43	Gemma4 E2B	63.71	28.96	14.58	51.85
44	NuExtract 2.0 8B	62.92	23.50	6.39	49.42
45	Qwen3.5 0.8B	57.44	26.98	6.65	46.63
46	MiniCPM-V 4.5	57.39	25.30	1.13	45.85
47	Gemma4 12B	53.62	30.82	13.16	45.39
48	GLM-4.6V Flash	47.28	44.52	22.17	45.19

370 PDFs; value F1 (%). All 47 published systems + DocQuorum. Equal scores share a rank.

Source: Maintained docs/results snapshot, 2 Oct 2026; scorer c8e5969. Stored outputs, not fresh inference. ExtractBench was used in development.

Results UNIKIE-BENCH

82.62 MACRO F1 / 1ST

#	System	Macro F1
1	DocQuorum (ours)	82.62
2	Gemini-3-Pro	82.37
3	Qwen3-VL-Plus	80.20
4	Qwen3-VL-8B	79.12
5	Qwen-VL-Max	77.77
6	GLM-4.1V-9B	69.91
7	GPT-4o	69.15
8	MiMo-VL-7B-RL	68.88
9	InternVL3.5-8B	67.70
10	GPT-5	67.59
11	MiniCPM-V4.5-8B	67.15
12	Claude-Sonnet-4.5	66.39
13	Kimi-VL-A3B	66.14
14	Gemma-3-12B	48.09
15	Ministral-3-8B	37.62
16	SmolVLM2-2.2B	23.69

4,472 images
11 document scenarios.

Six-reading quorum
A separate benchmark protocol, including extra-model readings.

This score is not attributed to the single-document API alone.

Source: Maintained docs/results snapshot: UNIKIE-BENCH Table 4 and stored DocQuorum evaluation. All 15 published systems + DocQuorum.